Non-Pseudo-Arithmetic Hyperbolic 4-Manifolds

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24 September 2026

Abstract

We give three explicit finite-volume hyperbolic Coxeter lattices in dimension four whose adjoint trace fields are not totally real. They are the reflection groups of the Ma–Zheng polytopes $P_{11,6}, P_{11,7}, P_{11,8}$. The obstruction is contained in two cyclic products of the doubled unit-normal Gram matrix $A = 2G$. Those products recover elements $s = \sqrt{5}$ and u satisfying

$$(s-1)u^2 + 2(s-3)u + s - 7 = 0.$$

After the embedding $s \mapsto -\sqrt{5}$, this quadratic has discriminant $8(1-\sqrt{5}) < 0$. The cyclic-product field therefore has a complex place. Vinberg's identification of that field with the adjoint trace field and the Emery–Mila total-reality obstruction imply that the three lattices are not pseudo-arithmetic. Torsion-free orientation-preserving finite-index subgroups yield orientable noncompact finite-volume hyperbolic 4-manifolds with the same obstruction.

Exact computations check the census data and provide independent field and finite-volume certificates. A pinned Lean 4 development verifies the radical identity and the number-field obstruction. It also checks the lattice and manifold implications conditional on explicit geometric and literature hypotheses.

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1 Introduction

1.1 The question and the answer

A *lattice* in the isometry group of real hyperbolic n-space is a discrete group whose quotient has finite hyperbolic volume. Arithmetic lattices are constructed from algebraic groups over number fields. Many important nonarithmetic lattices are assembled by gluing arithmetic pieces. Emery and Mila introduced *pseudo-arithmeticity* to give one common algebraic framework for arithmetic lattices, quasi-arithmetic lattices, and the familiar gluing constructions [2]. They asked whether every lattice in $\mathrm{PO}(n,1)$, for $n > 3$, is pseudo-arithmetic and, more weakly, whether every such lattice has totally real adjoint trace field [2, Questions 1.8 and 1.10].

Both questions have negative answers in dimension four.

Theorem 1.1 (Main theorem)**.** *For $j \in \{6,7,8\}$, let $P_{11,j}$ be the finite-volume hyperbolic Coxeter 4-polytope in the Ma–Zheng census recorded in Table 1, and let $\Gamma_{11,j} < \mathrm{PO}(4,1)$ be its reflection group. Then the adjoint trace field of $\Gamma_{11,j}$ is not totally real. Consequently $\Gamma_{11,j}$ is not pseudo-arithmetic.*

Corollary 1.2 (Manifold form)**.** *There exist orientable, noncompact, finite-volume hyperbolic 4-manifolds whose fundamental groups are not pseudo-arithmetic.*

Ma and Zheng classify all 331 finite-volume hyperbolic Coxeter 4-polytopes with seven facets [5]. The present examples are three consecutive entries in that census. The proof needs only four entries of their Gram matrices and one common radical. It does not require a numerical approximation, a primitive element for the entire field, or a classification of all embeddings of that field.

Emery–Mila checked the higher-dimensional Coxeter examples then available to them. The final 2026 version of Belolipetsky–Bogachev–Kolpakov–Slavich still described the known examples in $\mathrm{PO}(n,1)$, $n > 3$, as pseudo-arithmetic [1]. A targeted literature search through 24 September 2026 found no earlier recorded non-pseudo-arithmetic lattice in these dimensions. We know of no previous example.

1.2 Definitions and conventions

We recall the definitions used in the argument.

Hyperbolic space and lattices. On $\mathbb{R}^{n+1}$, put

$$\langle x, y\rangle_{n,1} = x_1y_1 + \cdots + x_ny_n - x_{n+1}y_{n+1}.$$

The upper sheet

$$\mathbb{H}^n = \{x \in \mathbb{R}^{n+1} : \langle x, x\rangle_{n,1} = -1,\ x_{n+1} > 0\}$$

is the hyperboloid model of real hyperbolic space. The subgroup of the orthogonal group preserving this sheet acts by isometries and is naturally isomorphic to $\mathrm{PO}(n,1) = \mathrm{O}(n,1)/\{\pm I\}$. A subgroup $\Gamma < \mathrm{Isom}(\mathbb{H}^n)$ is *discrete* if the identity is isolated in Γ. A discrete subgroup is a *lattice* if $\Gamma\backslash\mathbb{H}^n$ has finite volume. The quotient is an orbifold in general and a manifold when Γ is torsion-free.

Coxeter polytopes and Gram matrices. A convex hyperbolic polytope is *Coxeter* when every nonzero dihedral angle is π/m for an integer $m \geq 2$; zero angles at ideal intersections are also allowed. Reflections in its facets generate a discrete group for which the polytope is a fundamental domain. If the polytope has finite volume, this reflection group is a lattice. Choose an outward unit spacelike normal e_i to each facet. The matrix $G = (\langle e_i, e_j\rangle_{n,1})$ is the *unit-normal Gram matrix*. Its diagonal entries are one. Off the diagonal one has

$$g_{ij} = \begin{cases} -\cos(\pi/m_{ij}), & \text{the facets meet at angle } \pi/m_{ij}, \\ -1, & \text{the facets are parallel,} \\ -\cosh(\rho_{ij}), & \text{they are disjoint at distance } \rho_{ij}. \end{cases}$$

We use $A = 2G$. A *cyclic product* is a product $A_{i_0i_1}A_{i_1i_2}\cdots A_{i_mi_0}$. The field generated over $\mathbb{Q}$ by every such product is the *Vinberg field* $k(P)$.

Number fields and total reality. A *number field* is a finite extension of $\mathbb{Q}$. An embedding of a number field K is an injective field homomorphism $K \hookrightarrow \mathbb{C}$ fixing $\mathbb{Q}$. The field K is *totally real* when every such embedding has image in $\mathbb{R}$. Thus one nonreal embedding is a complete certificate that K is not totally real. We use the standard extension theorem: if $L \subset K$ are number fields, every embedding $L \hookrightarrow \mathbb{C}$ extends to an embedding $K \hookrightarrow \mathbb{C}$. This follows from the existence of a complex root of the transported minimal polynomial; a proof is recalled in Proposition 2.3.

Adjoint trace fields. For a matrix Lie group, conjugation differentiates to a linear action Ad on its Lie algebra. The *adjoint trace field* of a lattice Γ is

$$k(\Gamma) = \mathbb{Q}\big(\{\mathrm{tr}(\mathrm{Ad}\,\gamma) : \gamma \in \Gamma\}\big) \subset \mathbb{R}.$$

It is unchanged on passing to a finite-index subgroup. Vinberg's field-of-definition theorem identifies this field, for a finite-volume Coxeter reflection group, with the cyclic-product field $k(P)$. This identification is an imported theorem, not a new result proved here.

Pseudo-arithmeticity. Vinberg associates to a Zariski-dense lattice Γ an algebraic group over $k(\Gamma)$, called its ambient group. An algebraic group over a totally real field k is *admissible* for hyperbolic n-space when one real factor is the required noncompact projective orthogonal group and the other real factors are compact. In Emery–Mila's definition, a lattice is *pseudo-arithmetic* when its ambient group is obtained from such an admissible group by scalar extension through a finite totally real multiquadratic extension K/k. Their definition implies that $K = k(\Gamma)$ is totally real. This last necessary condition is the only part of pseudo-arithmeticity used in the proof.

Formal verification terminology. Lean checks a theorem only from its declarations, imported library theorems, and explicitly supplied hypotheses. In this release, the new algebraic argument is proved inside Lean. A literature result such as Vinberg's theorem is represented by a named hypothesis with the same logical type needed by the manuscript. Lean checks every implication after that boundary, but this does not constitute a new formal proof of Vinberg's theorem. No project-defined `axiom`, `opaque`, `sorry`, or `admit` is used.

1.3 The certificate in one paragraph

Let G be any of the three unit-normal Gram matrices and put $A = 2G$. With zero-based facet indices, the common entries

$$A_{12} = -\frac{1+\sqrt{5}}{2}, \qquad A_{03} = -1, \qquad A_{35} = -\sqrt{2}, \qquad A_{50} = -2a$$

give two cyclic products

$$c_{12} = A_{12}A_{21} = \frac{3+\sqrt{5}}{2}, \qquad c_{035} = A_{03}A_{35}A_{50} = -2\sqrt{2}\,a.$$

Hence the Vinberg field contains $s = 2c_{12} - 3 = \sqrt{5}$ and $u = -c_{035}/2 = \sqrt{2}a$. The exact radical for a implies

$$(s-1)u^2 + 2(s-3)u + s - 7 = 0.$$

At the conjugate embedding of $\mathbb{Q}(\sqrt{5})$, this quadratic has negative discriminant. It therefore produces a nonreal embedding of $\mathbb{Q}(s,u)$, which extends to the whole Vinberg field. This is the entire new obstruction.

1.4 Logical status of the ingredients

For clarity, the proof has four layers.

L1. *Elementary proof:* the matrix entries, two products, radical elimination, discriminant, and complex-place deduction are written out in Sections 4–6.

L2. *Exact computation:* independent Python and PARI/GP programs check the source transcription, rank, inertia, field signatures, and finite-volume combinatorics. They support the input data but are not needed to trust the two-cycle algebra once the displayed data are accepted.

L3. *Lean kernel checking:* the internal implication from the radical formula and cycle certificate to non-total reality and non-pseudo- arithmeticity is formalized. The finite-index corollary is checked from explicitly named geometric hypotheses.

L4. *External mathematics:* the census classification, Vinberg's field identification, the Emery–Mila necessary condition, Selberg's lemma, and finite-index invariance are cited rather than reproved. The exact list appears in Section 9.

This separation prevents an executable check of a matrix identity from being misdescribed as a formal verification of a classification theorem.

2 Basic field and hyperbolic facts

2.1 Embeddings, minimal polynomials, and complex places

We record the elementary field facts at the level needed later.

Definition 2.1. Let K be a number field and $\alpha \in K$. The *minimal polynomial* $\mathrm{minpoly}_{\mathbb{Q}}(\alpha)$ is the unique monic polynomial of least positive degree in $\mathbb{Q}[X]$ that vanishes at α.

The uniqueness follows from Euclidean division: if two monic polynomials of least degree vanish, their difference has smaller degree and also vanishes. The kernel of the evaluation homomorphism $\mathbb{Q}[X] \to K$, $p \mapsto p(\alpha)$, is therefore generated by this polynomial.

Lemma 2.2. *The number $\sqrt{5}$ is irrational and*

$$\mathrm{minpoly}_{\mathbb{Q}}(\sqrt{5}) = X^2 - 5.$$

The two embeddings of $\mathbb{Q}(\sqrt{5})$ over $\mathbb{Q}$ send $\sqrt{5}$ to $\sqrt{5}$ and $-\sqrt{5}$, respectively.

Proof. Suppose $\sqrt{5} = p/q$ for coprime positive integers p, q. Then $p^2 = 5q^2$, so the prime 5 divides p^2, hence divides p. Write $p = 5r$. Substitution gives $q^2 = 5r^2$, so $5 \mid q$, contrary to coprimality. Thus $\sqrt{5} \notin \mathbb{Q}$. It is a root of the monic quadratic $X^2 - 5$; irrationality rules out a linear minimal polynomial, so this quadratic is minimal. Any embedding preserves addition, multiplication, and rational numbers, and hence must send $\sqrt{5}$ to a root of $X^2 - 5$. The two roots are $\pm\sqrt{5}$, and both choices define an embedding of $\mathbb{Q}(\sqrt{5})$. □

Proposition 2.3 (Extension of embeddings)**.** *Let $L \subset K$ be number fields and let $\sigma : L \hookrightarrow \mathbb{C}$ be an embedding. Then σ extends to an embedding $\widetilde{\sigma} : K \hookrightarrow \mathbb{C}$.*

Proof. This is the standard embedding-extension theorem; we recall its mechanism. Because number fields have characteristic zero, K/L is finite and separable. By the primitive-element theorem there is $\theta \in K$ with $K = L(\theta)$. Let $p \in L[X]$ be the minimal polynomial of θ. Apply σ coefficientwise to obtain $p^\sigma \in \mathbb{C}[X]$. The fundamental theorem of algebra supplies a root $z \in \mathbb{C}$ of p^σ. The map

$$L[X] \longrightarrow \mathbb{C}, \qquad \sum_i a_i X^i \longmapsto \sum_i \sigma(a_i) z^i$$

has (p) in its kernel and therefore descends to a homomorphism $L[X]/(p) \to \mathbb{C}$. Identifying $L[X]/(p)$ with $L(\theta) = K$ gives the extension. A unital homomorphism from a field has zero kernel, so it is an embedding. □

Lemma 2.4 (Quadratic real-root test)**.** *Let $a, b, c \in \mathbb{R}$, with $a \neq 0$, and put $D = b^2 - 4ac$. The equation $ax^2 + bx + c = 0$ has a real solution if and only if $D \geq 0$.*

Proof. Multiplying the equation by $4a$ and completing the square gives

$$(2ax + b)^2 = b^2 - 4ac = D.$$

A real square is nonnegative, proving necessity. If $D \geq 0$, the two values $(-b \pm \sqrt{D})/(2a)$ are real and substitution shows that they are solutions. □

Proposition 2.5 (A nonreal subfield embedding persists)**.** *Let $L \subset K$ be number fields. If L has a nonreal embedding, then K is not totally real.*

Proof. Let $\sigma : L \hookrightarrow \mathbb{C}$ have image not contained in $\mathbb{R}$. Proposition 2.3 extends it to $\widetilde{\sigma} : K \hookrightarrow \mathbb{C}$. If the image of $\widetilde{\sigma}$ were contained in $\mathbb{R}$, then so would be the image of its restriction σ, a contradiction. □

2.2 The hyperboloid model and reflecting hyperplanes

Let $V = \mathbb{R}^{n+1}$ with the form $\langle\ ,\ \rangle_{n,1}$ defined in Subsection 1.2. A nonzero vector is spacelike, null, or timelike according as its squared Lorentzian norm is positive, zero, or negative. If e is a unit spacelike vector, then

$$H_e = \{x \in \mathbb{H}^n : \langle x, e\rangle_{n,1} = 0\}$$

is a totally geodesic hyperplane, and

$$r_e(x) = x - 2\langle x, e\rangle_{n,1} e$$

is the reflection fixing H_e pointwise. Indeed, $r_e(e) = -e$, r_e fixes the orthogonal complement of e, and direct expansion shows $\langle r_e(x), r_e(y)\rangle_{n,1} = \langle x, y\rangle_{n,1}$.

For a polytope described as the intersection of half-spaces $\langle x, e_i\rangle_{n,1} \leq 0$, the vectors e_i are outward normals. The inner products of pairs of normals give the Gram formulas in Subsection 1.2; these are the standard two-dimensional Lorentzian calculations for intersecting, asymptotic, and ultraparallel hyperplanes [6, Chapter 3].

Definition 2.6. A Coxeter matrix is symmetric, has diagonal entries one, and has off-diagonal labels $m_{ij} \in \{2, 3, \ldots\} \cup \{\infty\}$. Its Coxeter group is generated by involutions r_i with relations $(r_i r_j)^{m_{ij}} = 1$ when $m_{ij} < \infty$. For a hyperbolic Coxeter polytope, these generators are the geometric facet reflections.

The fact that the geometric reflection group is discrete and that the polytope is a fundamental domain is the Poincaré–Coxeter theorem. We use it as standard background. Finite volume of the fundamental domain is exactly finite covolume of the group.

2.3 From an orbifold to a manifold

Two elementary finite-index observations will be used. First, the kernel of the orientation character $\Gamma \to \{\pm 1\}$ has index at most two. For a group generated by hyperplane reflections, each generator reverses orientation, so the kernel has index exactly two. Second, a subgroup of finite index d gives a degree-d orbifold cover and multiplies volume by d. Thus finite volume is preserved in both directions.

The following external theorem removes torsion.

Theorem 2.7 (Selberg's lemma)**.** *Every finitely generated subgroup of $\mathrm{GL}_N(F)$, where F has characteristic zero, has a torsion-free subgroup of finite index.*

We cite Selberg's original result [7]. A Coxeter group with finitely many facets is finitely generated, and its reflection representation is linear over a characteristic-zero field, so the theorem applies. A torsion-free discrete group acts freely on $\mathbb{H}^n$: an element fixing a point lies in a compact point stabilizer, and discreteness makes the cyclic subgroup finite, hence trivial. The quotient is therefore a manifold. Passing inside the orientation-preserving subgroup makes it orientable. Finally, a finite cover of a noncompact finite-volume quotient is noncompact: if the cover were compact, its continuous image would be compact, contrary to noncompactness of the base.

3 Adjoint trace fields and pseudo-arithmeticity

3.1 The adjoint representation and ambient group

Let $\mathbf{G}$ be a real algebraic matrix group with Lie algebra $\mathfrak{g}$. Conjugation by $g \in \mathbf{G}(\mathbb{R})$ maps $X \in \mathfrak{g}$ to gXg^{-1}. This linear map is $\mathrm{Ad}(g)$. Its trace is independent of a choice of basis for $\mathfrak{g}$. For a Zariski-dense subgroup Γ, define

$$k(\Gamma) = \mathbb{Q}\big(\mathrm{tr}(\mathrm{Ad}\,\gamma) : \gamma \in \Gamma\big).$$

For lattices in a connected noncompact simple Lie group, Borel density gives the required Zariski density. Vinberg's theory produces a form of the ambient adjoint algebraic group over $k(\Gamma)$, unique up to $k(\Gamma)$-isomorphism [8]. We do not need its construction; only the field is used below.

We also use the standard finite-index invariance

$$[\Gamma : \Lambda] < \infty \quad \Longrightarrow \quad k(\Lambda) = k(\Gamma). \tag{3.1}$$

This is part of Vinberg's commensurability theory. One inclusion is immediate from $\Lambda \subset \Gamma$; equality is the nontrivial external statement and is listed as Input E6 in Section 9.

3.2 Pseudo-admissibility

An algebraic k-group $\mathbf{H}$ is *admissible for* $\mathrm{PO}(n,1)$ when k is totally real and, among the groups obtained from the real embeddings of k, exactly one has the required noncompact factor $\mathrm{PO}(n,1)^\circ$, while the others are compact. This is the usual arithmetic admissibility condition.

Definition 3.1 (Emery–Mila)**.** Let $n > 3$. A group over a number field K is *pseudo-admissible over* K/k if it is the scalar extension $\mathbf{H}_K$ of an admissible k-group through a finite, possibly trivial, totally real multiquadratic extension

$$K = k(\sqrt{d_1}, \ldots, \sqrt{d_t}).$$

A lattice $\Gamma < \mathrm{PO}(n,1)$ is *pseudo-arithmetic* when its ambient group is pseudo-admissible [2, Definitions 1.6–1.7].

The trivial case $t = 0$ includes quasi-arithmetic lattices. Total reality means that every embedding of K is real, as required in [2, Section 1.4]. Equivalently, the generators $d_i \in k^\times$ can be chosen totally positive, meaning $\sigma(d_i) > 0$ for every embedding $\sigma : k \hookrightarrow \mathbb{R}$. Merely requiring $K \subset \mathbb{R}$ at one embedding would not imply total reality.

Proposition 3.2 (Total-reality obstruction)**.** *If $n > 3$ and $\Gamma < \mathrm{PO}(n,1)$ is pseudo-arithmetic, then its adjoint trace field $k(\Gamma)$ is totally real.*

Proof. We use the necessary condition stated by Emery–Mila immediately before Question 1.10 in [2, Section 1.4]. In the notation above, an embedding $\tau : K \hookrightarrow \mathbb{C}$ restricts to a real embedding σ of k, and $\tau(\sqrt{d_i})^2 = \sigma(d_i) > 0$. Hence every generator has real image, so $\tau(K) \subset \mathbb{R}$. Since $K = k(\Gamma)$, the conclusion follows. □

The contrapositive is what we use:

$$k(\Gamma) \text{ not totally real} \quad \Longrightarrow \quad \Gamma \text{ not pseudo-arithmetic.} \tag{3.2}$$

3.3 Vinberg's cyclic-product field

Let $P \subset \mathbb{H}^n$ be a finite-volume Coxeter polytope with unit-normal Gram matrix $G = (g_{ij})$, and put $A = 2G$. For a closed sequence of facet indices $i_0, i_1, \dots, i_m, i_0$, its cyclic product is

$$A_{i_0 i_1} A_{i_1 i_2} \cdots A_{i_{m-1} i_m} A_{i_m i_0}.$$

Repeated indices are allowed, and reversing the cycle gives the same value because A is symmetric. Define

$$k(P) = \mathbb{Q}(\text{all cyclic products in } A).$$

Theorem 3.3 (Vinberg)**.** *If P has finite volume and $\Gamma(P)$ is its reflection group, then*

$$k(\Gamma(P)) = k(P).$$

This is the cyclic-product form of Vinberg's field-of-definition theorem [8]; see also [2, Section 4.1] and [4, Section 3.1]. It is an external input. The factor two fixes the normalization of the displayed cyclic products: starting from unit normals, Vinberg's diagonal-two convention gives $A = 2G$. The generated field is unchanged if one uses G, since a cycle of length ℓ is multiplied by the rational factor 2^ℓ.

Remark 3.4 (Entry field versus trace field). The field generated by all individual entries of G can be strictly larger than $k(P)$. The relevant rescaling invariance concerns the reflection coefficients $a_{ij} = 2\langle e_i, e_j\rangle / \langle e_i, e_i\rangle$, which equal A_{ij} for unit normals. Replacing e_i by $\lambda_i e_i$ changes a_{ij} to $(\lambda_j/\lambda_i)a_{ij}$, so the factors cancel around a closed cycle. Raw bilinear Gram entries do not have this rescaling rule. Thus a radical in an individual Gram entry need not lie in the adjoint trace field; the field inclusions below follow from explicit cyclic products.

4 The three Coxeter polytopes and their exact matrices

4.1 How the census vector is read

We number the seven facets $0, 1, \dots, 6$. The 21 unordered pairs are listed in lexicographic blocks:

$$\begin{aligned}(01, 02, 03, 04, 05, 06) \mid (12, 13, 14, 15, 16) \mid (23, 24, 25, 26)\\ \mid (34, 35, 36) \mid (45, 46) \mid (56).\end{aligned} \tag{4.1}$$

An integer $m \geq 2$ means dihedral angle π/m, the symbol 0 means parallel facets, and a letter means an ultraparallel pair. Thus the initial block $2232ab$ means

$$m_{01} = 2, \quad m_{02} = 2, \quad m_{03} = 3, \quad m_{04} = 2, \quad g_{05} = -a, \quad g_{06} = -b.$$

The relevant published rows are as follows. The displayed volume is the volume of the Coxeter fundamental polytope.

The assertion that these rows define finite-volume polytopes is external Input E1. Section 7 describes independent exact checks, including a direct finite-volume check for $P_{11,8}$.

polytope	Coxeter vector	cusps	volume
$P_{11,6}$	`2232ab 52322 2220 442 22 2`	1	$13\pi^2/864$
$P_{11,7}$	`2232ab 52323 2220 442 22 2`	1	$\pi^2/54$
$P_{11,8}$	`2232ab 52324 2220 442 22 2`	2	$19\pi^2/864$

Table 1: The three Ma–Zheng census rows used in the proof.

4.2 Radicals and a uniform matrix

Put

$$s = \sqrt{5}, \qquad r = \sqrt{2},$$

and let $q_6 = 2, q_7 = 3, q_8 = 4$. The common first dotted entry is

$$a = \frac{1}{2}\sqrt{7 + s + 2\sqrt{2 + 2s}}. \tag{4.2}$$

The second dotted entries are

$$\begin{aligned} b_6 &= \sqrt{\frac{9}{2} + 2s + \sqrt{38 + 17s}}, \\ b_7 &= r + \frac{\sqrt{10}}{2} + \frac{1}{2}\sqrt{29 + 13s}, \\ b_8 &= \frac{1}{4}\left(2 + 3r + 2s + rs + 2\sqrt{19 + 14r + 9s + 6rs}\right). \end{aligned} \tag{4.3}$$

All square roots mean the positive real root. Their radicands are positive, and the resulting a, b_j exceed one, as required for ultraparallel facets. Only a is used in the obstruction.

For reference, the complete unit-normal Gram matrix is

$$G_j = \begin{pmatrix} 1 & 0 & 0 & -\frac{1}{2} & 0 & -a & -b_j \\ 0 & 1 & -\frac{1+s}{4} & 0 & -\frac{1}{2} & 0 & -\cos(\pi/q_j) \\ 0 & -\frac{1+s}{4} & 1 & 0 & 0 & 0 & -1 \\ -\frac{1}{2} & 0 & 0 & 1 & -\frac{r}{2} & -\frac{r}{2} & 0 \\ 0 & -\frac{1}{2} & 0 & -\frac{r}{2} & 1 & 0 & 0 \\ -a & 0 & 0 & -\frac{r}{2} & 0 & 1 & 0 \\ -b_j & -\cos(\pi/q_j) & -1 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{4.4}$$

To check the transcription, read each row against (4.1). For example, $m_{12} = 5$ gives $g_{12} = -\cos(\pi/5) = -(1+s)/4$; $m_{26} = 0$ gives $g_{26} = -1$; and $m_{34} = m_{35} = 4$ gives $-r/2$. The elementary special-angle identities used here are

$$\cos(\pi/3) = \frac{1}{2}, \qquad \cos(\pi/4) = \frac{\sqrt{2}}{2}, \qquad \cos(\pi/5) = \frac{1+\sqrt{5}}{4}.$$

The first two follow from an equilateral triangle and an isosceles right triangle. For the third, put $z = e^{i\pi/5}$ and $x = z + z^{-1} = 2\cos(\pi/5)$. Since $z^5 = -1$ and $z \neq -1$, division of $z^5 + 1$ by $z + 1$ gives $z^4 - z^3 + z^2 - z + 1 = 0$. After division by z^2, this becomes

$$(z^2 + z^{-2}) - (z + z^{-1}) + 1 = 0.$$

Because $z^2 + z^{-2} = x^2 - 2$, we get $x^2 - x - 1 = 0$. Positivity selects $x = (1+\sqrt{5})/2$.

4.3 The two cyclic products

Put $A_j = 2G_j$. Four common entries of A_j are

$$A_{12} = -\frac{1+s}{2}, \qquad A_{03} = -1, \qquad A_{35} = -r, \qquad A_{50} = -2a. \tag{4.5}$$

The first closed path has two directed edges, $1 \to 2 \to 1$, and gives

$$\begin{aligned} c_{12} = A_{12}A_{21} &= \left(-\frac{1+s}{2}\right)^2 \\ &= \frac{1+2s+s^2}{4} = \frac{6+2s}{4} = \frac{3+s}{2}, \end{aligned} \tag{4.6}$$

where $s^2 = 5$. Thus

$$s = 2c_{12} - 3 \in k(P_{11,j}). \tag{4.7}$$

The second closed path is $0 \to 3 \to 5 \to 0$:

$$c_{035} = A_{03}A_{35}A_{50} = (-1)(-r)(-2a) = -2ra. \tag{4.8}$$

Consequently

$$u := ra = -\frac{1}{2}c_{035} \in k(P_{11,j}). \tag{4.9}$$

These equations prove membership in the cyclic-product field.

5 The radical identity and the complex place

5.1 Deriving the quadratic relation

We now prove the one algebraic identity behind all three cases.

Lemma 5.1 (Exact radical elimination)**.** *Let $s = \sqrt{5}$, let a be given by (4.2), and let $u = \sqrt{2}a$. Then*

$$(s-1)u^2 + 2(s-3)u + s - 7 = 0. \tag{5.1}$$

Proof. Set

$$t = \sqrt{2+2s} > 0.$$

By the definitions of a and u,

$$u^2 = 2a^2 = 2 \cdot \frac{1}{4}(7+s+2t) = \frac{7+s}{2} + t. \tag{5.2}$$

We claim first that

$$(s-1)u = 3 - s + t. \tag{5.3}$$

Both sides are positive: $s > 1$ and $u > 0$ for the left side, while $s < 3$ and $t > 0$ for the right side. It is therefore enough to prove that their squares are equal. Using $s^2 = 5$, $t^2 = 2+2s$, and (5.2), expansion gives

$$\begin{aligned} ((s-1)u)^2 &= (s-1)^2\left(\frac{7+s}{2} + t\right) \\ &= (6-2s)\left(\frac{7+s}{2} + t\right) \\ &= 16 - 4s + (6-2s)t, \end{aligned}$$

because $(6-2s)(7+s)/2 = (42-8s-2s^2)/2 = 16-4s$. On the other hand,

$$\begin{aligned}(3-s+t)^2 &= (3-s)^2 + 2(3-s)t + t^2 \\ &= 14 - 6s + 2(3-s)t + 2 + 2s \\ &= 16 - 4s + (6-2s)t.\end{aligned}$$

The positive quantities have equal squares, proving (5.3).

Rearrange it as

$$t = (s-1)u + s - 3. \tag{5.4}$$

Squaring, substituting $t^2 = 2 + 2s$, and moving everything to the left gives

$$((s-1)u + s - 3)^2 - (2 + 2s) = 0.$$

A direct expansion factors the left side as

$$(s-1)\big((s-1)u^2 + 2(s-3)u + s - 7\big).$$

Because $s \neq 1$, division by $s-1$ proves the claimed relation. □

Remark 5.2. The expanded proof uses only positivity of the named square roots, $s^2 = 5$, and ring arithmetic. The Lean theorem

`explicit_radical_cycle_relation`

checks the same identity from the nested square-root definitions. The intermediate theorem

`cycle_relation_of_radical_identities`

checks the final factorization over an arbitrary characteristic-zero field.

5.2 Negative discriminant at the conjugate place

Lemma 5.3 (Non-total-reality certificate)**.** *Let $s = \sqrt{5} \in \mathbb{R}$ and let $u \in \mathbb{C}$ satisfy (5.1). Then $L = \mathbb{Q}(s,u)$ is not totally real.*

Proof. View (5.1) as $f(u) = 0$, where

$$f(X) = (s-1)X^2 + 2(s-3)X + s - 7 \in \mathbb{Q}(s)[X].$$

Its leading coefficient is nonzero because $s \neq 1$. Its discriminant is

$$\begin{aligned}\Delta &= \big(2(s-3)\big)^2 - 4(s-1)(s-7) \\ &= 4\big((s-3)^2 - (s-1)(s-7)\big) \\ &= 4\big((s^2 - 6s + 9) - (s^2 - 8s + 7)\big) \\ &= 8(1+s).\end{aligned} \tag{5.5}$$

Let $\sigma : \mathbb{Q}(s) \hookrightarrow \mathbb{R}$ be the nonidentity embedding, so $\sigma(s) = -s$. Applying σ to the coefficients gives a real quadratic f^σ with discriminant

$$\sigma(\Delta) = 8(1-s) < 0, \tag{5.6}$$

because $s = \sqrt{5} > 1$. By Lemma 2.4, f^σ has no real root.

Since f is a nonzero polynomial with $f(u) = 0$, the extension $L = \mathbb{Q}(s,u)$ is a number field. By Proposition 2.3, σ extends to an embedding $\widetilde{\sigma} : L \hookrightarrow \mathbb{C}$. Applying this embedding to $f(u) = 0$ shows that $\widetilde{\sigma}(u)$ is a root of f^σ, hence is nonreal. Therefore L is not totally real. □

There is a shorter intrinsic formulation. If a number field K contains s, u with

$$\mathrm{minpoly}_{\mathbb{Q}}(s) = X^2 - 5, \qquad (s-1)u^2 + 2(s-3)u + s - 7 = 0,$$

then the root $-\sqrt{5}$ of the minimal polynomial is realized as the image of s under some embedding $K \hookrightarrow \mathbb{C}$. If that embedding were real, the image of u would solve the impossible conjugate quadratic. This is exactly the formulation kernel-checked in `not_totallyReal_of_cycle_minpoly`. For the positive radicals used here, $\mathbb{Q}(s,u) \subset \mathbb{R}$ in its distinguished embedding. The conclusion concerns its other embeddings: at least one has nonreal image.

6 Proofs of the main theorem and manifold corollary

Proof of Theorem 1.1. Fix $j \in \{6, 7, 8\}$. Let $\Gamma_j = \Gamma(P_{11,j})$ be the group generated by reflections in the facets of $P_{11,j}$. By the Ma–Zheng census statement, $P_{11,j}$ has finite volume. Therefore Γ_j is a lattice.

Let $K_j = k(\Gamma_j)$ be its adjoint trace field. By Theorem 3.3,

$$K_j = k(P_{11,j}). \tag{6.1}$$

The products in (4.6) and (4.8) are cyclic products of $A_j = 2G_j$. Equations (4.7) and (4.9) therefore show that

$$s = \sqrt{5} \in K_j, \qquad u = \sqrt{2}a \in K_j.$$

All entries of G_j are algebraic, so their entry field is a number field, as is its subfield K_j. Consequently $L = \mathbb{Q}(s,u)$ is a number subfield of K_j.

Lemma 5.1 gives the quadratic relation for s, u, and Lemma 5.3 gives a nonreal embedding of L. By Proposition 2.5, that embedding extends to a nonreal embedding of K_j. Thus K_j is not totally real.

If Γ_j were pseudo-arithmetic, Proposition 3.2 would make K_j totally real. This contradicts the preceding paragraph. Hence Γ_j is not pseudo-arithmetic. The computation was independent of j, so it applies to all three rows. □

Proof of Corollary 1.2. Choose one j and write $\Gamma = \Gamma_{11,j}$. Let $\Gamma^+ = \Gamma \cap \mathrm{Isom}^+(\mathbb{H}^4)$. The orientation character shows that $[\Gamma : \Gamma^+] = 2$. As a finite-index subgroup of the finitely generated reflection group, Γ^+ is finitely generated; it is also linear in characteristic zero. By Selberg's lemma, it contains a torsion-free subgroup Λ of finite index.

The action of Λ on $\mathbb{H}^4$ is free and orientation preserving, so

$$M = \Lambda \backslash \mathbb{H}^4$$

is an orientable hyperbolic 4-manifold. Since Λ has finite index in the lattice Γ, its quotient has finite volume. The base orbifold is noncompact because the census row has at least one ideal vertex; the finite cover M is therefore noncompact as explained in Subsection 2.3.

Finite-index invariance of the adjoint trace field gives

$$k(\Lambda) = k(\Gamma) = K_j.$$

Theorem 1.1 says that this field is not totally real. Applying Proposition 3.2 to Λ shows that Λ, and equivalently the manifold with this fundamental group, is not pseudo-arithmetic. □

7 Exact computational certification

The preceding proof is hand-checkable after accepting the source rows and the two external field theorems. The repository also contains redundant exact checks. Their purpose is to detect transcription mistakes and to make every concrete assertion reproducible.

7.1 Rank and inertia of all three Gram matrices

Exact symbolic elimination gives

$$\operatorname{rank}(G_j) = 5, \qquad \operatorname{inertia}(G_j) = (4, 1, 2)$$

for $j = 6, 7, 8$. Here the entries count positive, negative, and zero eigenvalues. The common principal block on facets $0, 1, 2, 3, 4$ has leading principal minors

$$1, \quad 1, \quad \frac{5-s}{8}, \quad \frac{3(5-s)}{32}, \quad -\frac{1+s}{32}. \tag{7.1}$$

They are all nonzero because $1 < s < 3$. The signs are $+, +, +, +, -$; the Jacobi signature criterion therefore gives four positive and one negative direction on this block. Exact elimination shows that the full matrix has rank five, so the remaining two directions are null. The programs additionally verify that every principal six-by-six minor vanishes. The nonzero five-by-five minor and the elimination certificate are the rank witnesses; no floating-point eigenvalue is used.

7.2 Direct finite-volume check for the third polytope

For $P_{11,8}$, Lean classifies all 127 nonempty principal submatrices: 54 are positive definite, 12 are positive semidefinite but not positive definite, and 61 have a negative direction. It proves that exactly ten subsets are maximal positive semidefinite, matching the vertex incidences of the published combinatorial type. Eight vertices have elliptic rank-four links. The remaining two are parabolic:

$$\{1, 3, 4, 6\} \quad \text{of type } \widetilde{C}_3, \qquad \{2, 3, 4, 5, 6\} \quad \text{of type } \widetilde{A}_1 \sqcup \widetilde{C}_2.$$

The second has nullity two and is the Euclidean triangular-prism link of a nonsimple ideal vertex. Every elliptic rank-three edge diagram has exactly two ordinary or ideal endpoints. Vinberg's finite-volume criterion [9] then gives finite volume and exactly two cusps.

An independent CoxIter run, pinned to a recorded source commit, returns

$$(f_0, f_1, f_2, f_3, f_4) = (10, 21, 18, 7, 1), \qquad \operatorname{vol}(P_{11,8}) = \frac{19\pi^2}{864},$$

agreeing with the census. Lean independently proves all five face counts by closing the vertex–facet incidences and counting nonempty faces, including the polytope, by strict-chain dimension. The correspondence between this incidence poset and geometric faces uses the published realization; no numerical face count is assumed in the Lean calculation. CoxIter is described in [3]. These checks verify the geometric input for one case; the proof of the family still cites the Ma–Zheng classification for all three.

7.3 A complete trace-field check for the third polytope

The two cycles used in the theorem give only the subfield needed for the obstruction. Two further cycles recover $r = \sqrt{2}$ and b_8:

$$c_{126} = -r(1+s), \qquad c_{03416} = -4b_8.$$

Since every Gram entry lies in $\mathbb{Q}(s, r, a, b_8)$, this proves

$$k(P_{11,8}) = \mathbb{Q}(s, r, a, b_8).$$

With $w = 2(r + a + b_8)$, exact power-basis reconstruction gives $k(P_{11,8}) = \mathbb{Q}(w)$, where

$$\begin{aligned} p_w(X) = X^8 - 8X^7 - 252X^6 - 1392X^5 - 3524X^4 \\ - 5280X^3 - 5712X^2 - 3968X - 1136. \end{aligned} \tag{7.2}$$

PARI/GP verifies irreducibility, signature $(r_1, r_2) = (4, 2)$, and discriminant

$$40960000 = 2^{16}5^4.$$

Thus the full field independently exhibits two conjugate pairs of complex embeddings. A second primitive-element route produces a different polynomial for the same explicitly reconstructed field; agreement is not inferred from degree, signature, and discriminant alone.

7.4 Artifacts and bounded-memory execution

The exact programs and durable outputs are:

- `scripts/p11_6_7_non_total_trace.py`; `results/p11_6_7_non_total_trace_certificate.json`;
- `scripts/p11_8_non_total_trace.py`; `results/p11_8_non_total_trace_certificate.json`;
- `scripts/p118_trace_field.py`; `results/p118_trace_field_certificate.json`;
- `scripts/p118_field_independent.gp`, the independent PARI/GP irreducibility, real-root, signature, and discriminant check;
- `scripts/p11_8_finite_volume.py`; `results/p11_8_finite_volume_validation.json`;
- the pinned CoxIter input, output, and provenance files under `results/`.

Each expensive symbolic case runs in a fresh process. On the reference machine the case-worker resident-set peaks are below 100 MB. The exact commands, source URLs, versions, and expected digests are in `REPRODUCIBILITY.md`.

The Ma–Zheng source file used by the certifiers has SHA-256

`0017931637f017641d4935450cc2a07d070866d9c749443d9e3ede9f3a39f003`.

Each headline Python certifier checks this digest before parsing the rows.

8 Lean formalization and trust boundary

The directory `formal/` is a Lean 4 project pinned to Lean and mathlib `v4.32.0`. This section states exactly what its successful build means.

8.1 Unconditional kernel-checked mathematics

The modules `NonTotalReality.lean` and `Manuscript.lean` prove the following.

1. `recover_s_from_twoCycle` and `recover_u_from_threeCycle` check the rearrangements in (4.7) and (4.9) over every characteristic-zero field.

2. `explicit_radical_cycle_relation` proves (5.1) from the positive nested-radical definitions over $\mathbb{R}$.

3. `minpoly_eq_of_square_five` proves that $s^2 = 5$ implies $\mathrm{minpoly}_{\mathbb{Q}}(s) = X^2 - 5$ in any characteristic-zero field. It uses a proof of the irreducibility of $X^2 - 5$ over $\mathbb{Q}$.

4. `conjugateQuadratic_no_real_root` proves that the conjugate relation has no real solution. The theorem `not_totallyReal_of_cycle_minpoly` then uses mathlib's number-field embedding theorem to realize $s \mapsto -\sqrt{5}$ and deduce non-total reality. Its square-identity variant is `not_totallyReal_of_cycle_square`.

5. A `TwoCycleCertificate` records four field elements and four equations: the two cycle values, the minimal polynomial of s, and the relation for u. The constructor `of_real_cycles` builds this certificate from a real embedding $\iota : K \hookrightarrow \mathbb{R}$ and the two equalities

$$\iota(c_{12}) = (3 + \sqrt{5})/2, \qquad \iota(c_{035}) = -2\sqrt{2}a.$$

   It recovers $s, u \in K$, transports the radical relation by injectivity, and proves the minimal-polynomial statement. The theorem `not_totallyReal_of_real_cycles` concludes that K is not totally real.

The module `GramAudit.lean` also defines all three displayed matrices over $\mathbb{R}$, their entry fields, and their cyclic-product fields. It proves the cycle values, algebraicity of the entries, the number-field instances, and non-total reality of all three cyclic-product fields. For the third matrix it also proves equality of the cyclic-product and entry fields. These results require no geometric predicates. Identifying these matrices with the census polytopes, and their cyclic-product fields with the adjoint trace fields of the reflection groups, remains external.

8.2 Conditional geometric implications

The module `ExternalContracts.lean` specializes the imported statements to hyperbolic dimension four. Its structure `CensusInputs` records:

- a reflection group Γ_j for each of the three cases, related to its actual unit-normal matrix G_j;
- finite volume and noncompactness of its quotient as separate inputs;
- Vinberg's equality between the group's adjoint trace field and the field generated by closed products in $2G_j$; and

- the implication from pseudo-arithmeticity of a finite-volume lattice to total reality of its adjoint trace field.

The theorem `census_main_theorem` combines these inputs with the matrix-derived field proofs to conclude, for every case,

$$\text{finite-volume lattice} \ \wedge \ k(\Gamma_j) = k(G_j) \ \wedge \ \neg\text{total reality} \ \wedge \ \neg\text{pseudo-arithmeticity}.$$

Here $k(G_j)$ denotes the concrete cyclic-product field. The trace-field identification is a literal equality of intermediate fields of $\mathbb{R}/\mathbb{Q}$. The theorem requires no separately chosen field or two-cycle certificate. The group and manifold vocabulary has its cited geometric interpretation; Lean checks the implications from the stated contracts.

The structure `CoverInputs` requires one subgroup–quotient pair (Λ, M) satisfying finite index, torsion-freeness, orientation, and quotient correspondence simultaneously. Finite-index invariance is the equality $k(\Lambda) = k(\Gamma_j)$. The field `quotient_geometry` uses both finite volume and non-compactness of the base, and supplies the lattice property of Λ as well as the required manifold geometry. Thus the pseudo-arithmetic implication applies to the cover group with its hypotheses intact. The theorem `census_manifold_corollary` proves Corollary 1.2 from these contracts. The earlier interfaces `CoxeterExternalInputs` and `ManifoldExternalInputs` remain reusable abstract implications.

8.3 Statement-by-statement semantic audit

The register `audit/manuscript_statements.json` maps the mathematical assertions in every source paragraph, displayed equation, and table to their Lean declarations or to an explicit external boundary. Its readable companion is `notes/manuscript_semantic_audit.md`. Each entry records a positive example and a negative control: a failed hypothesis, a boundary case, or a counterexample to a proposed stronger statement. A valid theorem has no counterexample satisfying all its hypotheses. The register assesses translation separately from proof provenance: external inputs are acceptable when their specialization, hypotheses, and use match the text. Reusing a checked library result or a different proof is also acceptable.

The audit modules additionally prove the two embeddings of $\mathbb{Q}(\sqrt{5})$, embedding extension, the discriminant criterion, the equivalence of total reality and total positivity for nonzero defining square-root radicands, the Lorentz reflection identities, and elementary orientation-index and non-compactness statements. Exact scripts test all three matrices and all 127 principal subdiagrams of the third, and reject perturbed cycle values, power-basis coordinates, ranks, and vertex lists. The five displayed leading-minor identities, the complete principal-subdiagram classification for $P_{11,8}$, and its face vector are also proved in Lean. Further Lean controls reject all 70 single-bit vertex–facet incidence changes and all ten vertex deletions. They also prove that the incorrect vectors $(10, 21+t, 18+t, 7, 1)$, for every integer $t > 0$, pass the Euler alternating-sum check. The contract controls reject an unrelated field, a wrong case–matrix pair, a reversed implication, and incompatible existential witnesses. Logical test models do not instantiate hyperbolic lattices or manifolds; the geometric interpretation remains the explicit imported input.

8.4 Other formal modules

The module `NonTotalReality.lean` contains the core field theorem in a minimal form. The separate module `ConditionalDescent.lean` concerns the supplementary normalized-Clifford and split-support criterion documented in the repository notes. It represents ABHN, local Brauer functoriality, Hasse–Minkowski, and the pseudo-arithmetic candidate classification as ordinary hypotheses

and checks their downstream implication graph. That criterion is not needed for Theorem 1.1; the main examples are settled by non-total reality before any Clifford invariant is considered.

8.5 Audit statement

A release audit searches all project Lean sources for the declarations `axiom` and `opaque` and for the proof escapes `sorry`, `admit`, and `native_decide`. None occurs in project declarations or proofs; dependency sources are excluded from this textual check. A successful Lean build checks the proof terms against the kernel and imported library declarations. The theorem-level axiom audit reports only Lean's standard `propext`, `Classical.choice`, and `Quot.sound`. The geometric conclusions remain conditional on the hypotheses described above. The build command is

```
cd formal && lake build.
```

9 External-input register

This register records the geometric, field-theoretic, and library inputs. Each item distinguishes facts proved in the text or in mathlib from those assumed by the manuscript-facing Lean theorems.

E1: Census rows and finite volume.
Ma–Zheng's classification supplies the three vectors, positive radical values, cusp counts, volumes, and the assertion that they are finite-volume hyperbolic Coxeter 4-polytopes [5]. Exact scripts verify the transcription, Gram rank, and inertia; a direct vertex-link proof independently verifies finite volume for $P_{11,8}$. The Lean input `census_realization` relates each group to its actual matrix, while `finite_volume` and `noncompact` give the two separate geometric properties. The census rows themselves remain external data.

E2: Vinberg field identification.
For a finite-volume Coxeter polytope, the field generated by cyclic products in $2G$ is the adjoint trace field of the reflection lattice [8]. The Lean input `vinberg` requires the reflection group–matrix correspondence and finite volume, and concludes equality with the concrete cyclic-product field. The theorem `traceField_eq_cycleField` specializes it to each census case.

E3: Pseudo-arithmetic fields are totally real.
Emery–Mila's definition and Section 1.4 imply that a pseudo-arithmetic lattice in dimension $n > 3$ has totally real adjoint trace field [2]. The structure field `pseudoArithmetic_totallyReal` records this implication for finite-volume lattices in the fixed dimension four; Lean checks its use.

E4: Extension and realization of number-field embeddings.
The standard extension theorem and mathlib's theorem describing evaluations of number-field embeddings by roots of a minimal polynomial are used to realize $s \mapsto -\sqrt{5}$. The elementary extension proof is included in Proposition 2.3; Lean uses mathlib's existing verified result rather than a project hypothesis.

E5: Selberg's lemma.
A finitely generated characteristic-zero linear group has a torsion-free finite-index subgroup [7]. It is not formalized from first principles here. The Lean input `cover_exists` packages

the needed specialization after passing to the orientation subgroup, together with the quotient existence used in E7. All properties use the same witnesses.

E6: Finite-index trace-field invariance.
The adjoint trace field is a commensurability invariant [8]. The input `traceField_invariant` requires a finite-volume base lattice and a finite-index subgroup, and concludes equality of their field values.

E7: Finite-cover quotient geometry.
The standard facts that a torsion-free orientation-preserving subgroup gives an orientable manifold, finite index preserves finite volume, and a finite cover of the noncompact quotient remains noncompact are proved at the elementary level in Subsection 2.3. Lean packages their geometric realization as `quotient_geometry`, including the cover's lattice property. The input `quotient_pseudoArithmetic` relates the same group and manifold. These are specialized external statements; the project does not construct hyperbolic orbifolds.

No item E1–E7 is hidden behind a global Lean axiom. Replacing an input by a future library theorem would change the constructor used to build the corresponding structure, not the checked proof downstream.

10 Reproducibility and release audit

The repository is designed so that the central conclusion can be checked in three complementary ways.

Level 1: read the certificate. Check the four entries in (4.5), multiply the cycles, follow Lemmas 5.1 and 5.3, and apply Inputs E2 and E3. No computer algebra is required.

Level 2: build the formal and written proofs. Run

```
cd formal
lake build

cd ../manuscript
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  non_pseudo_arithmetic_4_manifolds.tex
```

The project pins Lean/mathlib, and the TeX source has a fixed date.

Level 3: repeat all exact source checks. Follow `REPRODUCIBILITY.md` to obtain the pinned Ma–Zheng source and HCPdm incidence file, verify their SHA-256 values, and run each Python or PARI/GP job serially. Reference JSON certificates are committed, so the mathematical outputs remain inspectable without network access. Timing and peak-memory metadata may vary and are not used as mathematical evidence.

The command

```
python3 scripts/release_check.py
```

performs the local release audit: JSON parsing, Python byte-compilation, forbidden-token checks, documentation consistency, Lean build, exact regressions, and TeX build. The packaging script `scripts/build_release.sh` creates a deterministic archive and checksum under `dist/`. The execution record is split between `PRE_RELEASE_CHECKLIST.md` and `RELEASE_NOTES.md`.

A Dependency graph of the main conclusion

For quick auditing, the complete argument is the following chain.

$$
\begin{array}{c}
\text{Ma–Zheng row (E1)} \\
\Downarrow \\
\text{four entries of } A = 2G \\
\Downarrow \\
c_{12} = (3+s)/2, \quad c_{035} = -2u \\
\Downarrow \\
s, u \in k(P), \quad \operatorname{minpoly}_{\mathbb{Q}}(s) = X^2 - 5 \\
\Downarrow \\
(s-1)u^2 + 2(s-3)u + s - 7 = 0 \\
\Downarrow \\
\text{an embedding extending } s \mapsto -\sqrt{5} \text{ sends } u \text{ outside } \mathbb{R} \\
\Downarrow \\
k(P) \text{ is not totally real} \\
\Downarrow \text{ Vinberg (E2)} \\
k(\Gamma) \text{ is not totally real} \\
\Downarrow \text{ Emery–Mila (E3)} \\
\Gamma \text{ is not pseudo-arithmetic.}
\end{array}
$$

The manifold corollary appends E5–E7. The radical relation and both downstream implication chains are checked by Lean; E1–E3 and E5–E7 remain visible external inputs.

B Independent full-field data for comparison

The full-field computation is not part of the proof, but it gives a useful cross-check. The signature $(4, 2)$ of the degree-eight field means that it has four real embeddings and two conjugate pairs of nonreal embeddings, since $r_1 + 2r_2 = 8$. The two-cycle proof detects at least one of those pairs without constructing the other six embeddings. The short certificate therefore proves the required obstruction without a primitive-element calculation. The full-field computation gives the additional degree, signature, and discriminant data.

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